

DSA 8405 Probability and Stochastic Processes-20250911_170625UTC-Meeting Recording

September 11, 2025, 2:30PM

2h 36m 19s



Odhiambo, Collins 4:17

Good evening, everyone. We're going to start shortly. Just give me a shot. I mean, you're going to start shortly.

OK, so we're going to.

We're going to start or rather continue from where we stopped last time. We started the whole concept of modelling queuing systems and so we took.

A bit of theory we took look we we we focus on a bit of theory before we can start at the mathematical part of it.

So last week we looked at queuing systems and a bit of components of a queuing system and the whole concept of theory of queue queuing theory, the structures of queuing systems.

And a bit of modeling.

OK, so we talked of the components that make a queue. A queuing system is characterized by three components. So there's the arrival process. So items have to arrive or entities have to arrive.

This can be depending on what you're modeling. Can be people, can be items, can be aeroplanes. If you're modeling air traffic control, it can be.

In in ICT it can be CPU time slice, so it depends on what you are looking at. In telecommunication it could be the calls that arrive at customer service. So there there's the arrival component of items or entities.

And then naturally, the entities will form a queue. Um.

So as long as the rate at which service is done is lower than the rate at which items arrive, then the items will naturally form a queue. So we'll have what we call a queuing display.

So the way queues are arranged is what we call a queuing display, and we did look at different types of queuing display.

Like FIFO, LIFO, 0, priority queue and all that. So some of the common queuing discipline that exist.

And then we also looked at service mechanism as a component of a queuing system. Service mechanism represents the time it takes to service a customer who is waiting

and service can be many things. It can be CPU time slice, it can be connection. Over a telephone call, it can be shooting down an enemy in a military war, so it can be anything and that service time is random. OK, so the arrival pattern is random the way items arrive.

To the system is arrival is random and then depending on the queuing discipline, then the service time, the time it takes to service the the item or whatever is on the queue is also random and that one is what we call a service mechanism.

When we are modelling A queuing system, we need to look at the nature of these three components because they are the ones which will define eventually a queue.

So we say that a queuing theory is a collection of mathematical model of various queuing systems that can take as input parameters from arrivals and then the queuing discipline and then provide and then also quantity and then the.

Service mechanism and from those parameters we can provide quantities that describe the system performance. Remember the ultimate goal modeling is how it is to be efficient. So when we are doing modeling.

This type of queue or this type of model is how efficient you want to be. So this is a branch is is just one area of stochastic processes. So we've looked at random walk. We have looked at gambler swing, we have looked at counting processes and Poisson is one of the counting processes. So this queuing system is also one branch of stochastic processes.

So there are many stochastic processes as you later know, and so last week we spent a bit of time just looking at the whole concept of queuing system.

And of course the structures of skewing systems, we did look at that and then when it comes to modelling queues, that's where we stopped last time. We did not look at the mathematical bit of it. So today we're going to look at the maths, the.

Semactical bit of bit. Um.

OK, so.

So I will let me see if I can pull my whiteboard figure.

We.

OK, so let me explain this setting and then we can move to the whiteboard and see how we model very, very simple queues and how that works.

So the system. First of all, this is a stochastic process, and a stochastic process must have two.

2 aspects. So there's the state space and there's a parameter space and we've said that parameter is basically time. In most places in time is time, but it can be different

things and we saw like in.

Random walk that parameter is a number is a steps. The person can either take step going forward or backward and we saw in gambler screen that parameter is a number of games the gambler is playing. OK so so so but the state is a value.

The gambler of interest has the money that person has if you're looking at the gambler's ruin. OK, so if you're looking at the random walk, the position where the person is or the person taking the random steps is that's the state.

That's the state space, but now that is ordered by some parameter steps that it takes.

OK, so the place where is is this the state. So in a gambler's ruin you will talk of the value, the amount of money.

A A has has the state.

But the number of games are the parameters OK.

So here also we are instead interested with the state, the system state because that's that's a stochastic process. A stochastic process must have those two for you to qualify to be a stochastic process that what makes.

Stochastic processes are a bit advanced in in terms of mathematical approach because you have to consider the ordering. It's not. It's not just a typical data analysis where you collect data like if I want to study.

The age of students in this class, so I'll just go and collect that data. Each of you will give me their age, so somebody will be 28, another 13132. So I'll have all variety of age and then I'll get the average.

And after that I can go ahead and get a standard deviation of your age and then I can do other statistics like kurtosis or or skewness or whatever type of data I can I want to analyze about your age.

That is a static data I've just collected and I'm analyzing data that is at that state.

OK, but if I'm interested that with something that is happening, that data is collected, but it's also ordered. There's some ordering of that data, then it becomes stochastic.

There's there's a bit of ordering of that data.

OK, like the number of accidents that occur along Langata or whatever Bagadi Bagadi Rd. So the amount of accidents, the number of accidents that occur.

So if the whole of January I have 20, the whole of February I have 6, the whole of March along that road I have 25. So I'm dealing with a collection of random variables because the number of accidents.

Is a state is there is a is a random variable. I don't know how many, yeah, I don't know how many access will occur next month. So it's a random variable. It's a

measure that is random, but that random variable is attached with some ordering. So there's ordering of the month, the month where we collected that data.

So because we have parameters, we have the data itself, the number of accidents and then we have another parameter of which is ordering the month at which the accidents occur. So I have month 120 month 2-3. So then that's now becomes a stochastic process and.

I can now model either Poisson using Poisson or I can model the number of accidents using whatever types of Poisson or whatever type because that is a count. We call it count data that is count counting processes.

OK, so that's the idea of stochastic process. A stochastic process, generally you talk of the state space and then you have the parameter space. So the parameter space can be time or it can be any other parameter.

But usually it's time. OK, so a system is a number of customers. A system state is the number of customers in the system. It is a non-negative integer number. You cannot have -2 customers or -3.

OK. So just figure out that this, this is the number of customers in a bank, in a bank hall, OK, like equity bank along Moya Ave. over.

So, so if it is a bank like equity bank, the number of customers inside that bank call and those customers have come to be serviced or to get some services of certain of certain sort, then that is the number of customers in that bank is the.

System state that's is the number of customers in that system. So that bank we are calling that a queuing system. OK. So the number of customers is a system state.

System state is the number of customers in the system.
and it is non-negative.

So if you let PNT be the probability that there are N customers in the system at time T, because remember there are customers who are coming and others are going. So every time you count the number of customers in the system.

That number of customers in the system is a random number.

That number of customers is a random number.

There are those who are coming and there are those who are going. There are those who are coming. So between 8:00 to 9:00, any the number of customers in the system could be 30. Between 9:00 to 10:00, the number of customers in the system could be 50.

Because some other people have come, some people have left, so at any time that number is a random number.

At time T at a certain time T . So PNT is a probability that there are N customers in the system at time T .

Then this process, this process can be presented as a step transition diagram and I think we will look at that. I'll I'll, I'll I'll explain that on my whiteboard. So an arrival transforms the system from current state N to the next state $N + 1$.

And anytime there's an arrival, somebody arrives that transforms the system from step N to step $N + 1$ because the number of customers is the state of the system, is the system state.

And then the transition rate from a lower to a higher rate, it's the arrival therefore is λ . I'll I'll, I'll put it in perspective. Similarly, as customers are served by the server, the system undergoes transition.

From higher state to lower state. OK, so a process often referred as departures or deaths. So the rate at which this departures of car is what we call service rate μ because we are believing that once somebody is served, that person goes doesn't. That doesn't stand there.

You leave. Immediately you're served. You leave.

And of course, we are interested in coming up with some basic performance measures. We will work on that today given the transit.

Well, looks like I disappeared.

I think I'm back. Can somebody confirm if I'm back?

Can you hear me?

 **Straton Amodora** 22:45

Yes, we can do.

 **Elias Shamalla** 22:45

Yes.

 **Odhiambo, Collins** 22:47

OK, looks like I disappeared. I don't know.

All right, so we did, we did look at this is what we call the kindle way of classifying queueing system. So one of the basic way of classifying classification system classification is what we call.

MM one that is the simplest classification, but you may have different queues with with different ways of of how they look. If you read different books, you may find

different type of queues, but generally queues are classified using this kind of modification.

So when somebody's classifying the queue or when you're building the model for queue, you need.

Discussed at length, really that's for his own. You need to have service mechanism which we have also looked at and that is usually follows an exponential distribution. You you need to know the number of servers in that queue, how many servers are there in the queue.

You need to know the queuing B screen.

Is it following FIFO or LIFO? You need to know the capacity of the system. Is it unlimited queue or is it finite or infinite?

HM can hold and then you need to to define the understand the population size like where customers are coming. Is it a finite number or it is an infinite number? So this is.

This guy called Kendall is the one who came up with this classification and up today we use that methodology to come up with the queue. But now where we have this is the simplest type of queue.

This is the simplest where we assume the arrival is poisson. And then of course the population from where the arrivals originate is infinite. You cannot exhaust the population from where the arrival is coming from.

There's no limit in terms of the queue length and there's no restriction in terms of the queue length. The queue can grow Infinity and then there's no bulking. I explained bulking from the queue. What it means is that once you join the queue, you have to be served so you cannot exit or move ahead of.

Of the other person. So the the the queuing discipline will be strictly there too, and it will be FIFO.

So one object cannot arrive, one object at most arrive and then departs at an interval and then of course the service mechanism follows the exponential distribution. So this is the simplest type of queue and this is what we will model.

OK, so that's what we will model. So let me let me move away from this and just try to take us to my whiteboard.

OK, uh.

It's so.

OK, so let's see how we can do the mathematics now, the modeling of the queue. So we what we are saying is that I let.

Uh P.

NT.

It should be P.

And yeah, this should be two down here.

Hey, so there should be.

Mm.

OK, so this should be OK, so that's that's fine. So this is is a probability that there are N entities in the system.

In the system at at time T.

OK, so that's the probability that there are N entities in the system at time.

OK.

Let's just um, let's just write that as uh.

Um.

Yes, it's just like in.

OK, just write that as PN. Just write that just PN. OK, so that probability that there are N customers at time T, let's just write that as PN.

Now at steady state, so we consider so that this is meaning this means.

Um.

I.

Can you hear me guys, I think.

SA **Straton Amodora** 31:50

Yes, we can.

OC **Odhiambo, Collins** 32:11

I don't think if my net to uh getting on and off um.

I I'm OK back and today it's a bit awkward. I don't know why it's system is just.

Cutting me on and off.

So can somebody confirm you can hear me and you can see the whiteboard?

SA **Straton Amodora** 33:17

Yes, you can see. You can. Yeah, you can see the white.

JM **Jackline Mboya** 33:20

We can't hear you.

OC Odhiambo, Collins 33:21

OK. Thanks.

OK. Thanks. Um.

So just think it look at it this way.

In the morning they are and the rate of arrival is Λ . OK, so they are arrivals into the system and the rate of arrival is Λ .

It's random and then after after within the same hour there are also exits. People are being served. So within the same hour there's also an exit which is new.

The rate of meal is the rate of service. So once somebody served is moving out of the system. So that system within one hour there are people arriving, let's say, assume it's a bank call.

There are people arriving in the bank at random and the rate at which they are arriving is Λ OK and then because it's a point arrival is exit and the rate at which people exit.

SA Straton Amodora 35:08

I think you are lost.

OC Odhiambo, Collins 35:12

I'm lost again.

I don't know what's happening today. Uh.

The system is just.

On and off.

OK. So again at any time we have arrivals and then that is at the second hour and then we have exit for the second hour.

Then within the next three, on the third hour we have arrivals.

And then we have exits.

OK, so.

Customers are arriving into the bank hall and then they also served and they are exiting. So we have arrivals and then we have. So when the bank opens in the morning at 9:00, so between 9:00 and 10:00 we had.

Customers arriving and then within the same hour there was those customers that left. So the number of people inside here within that hour is a random number.

And that's what we are calling a stochastic process. And because it has arrival, then

there's a queuing discipline and then there are service where people come to get service and they exit. That's now what we call a queuing model, a queuing system. So you have stochastic class of systems where you have arrivals and then you have queues and then you have exits. So between 9:00 to 10:00 that number is a random number that is inside that form that is the.

State of the system. The state of the system is the number of entities inside the system in the system. OK, so again in.

They saw another there are those who are still there, but then they who are still in line and they were served and they exit. So the number amount of customers, the number of customers between 9:00 to 10:00 is a random variable and also between 10:00 to 9:00 is a random variable.

And then you go that way, you also find between 11:00 to 12:00 is also a random variable.

So, so, so.

It be the entity becomes a random and that is ordered with the time you put in between.

And that becomes a stochastic system and it's a class of stochastic process which we call the queuing model. So you have to figure it that way and then.

Um, consider, consider what we call a steady state.

OK, so a system is state to be in steady state if the rate of arrival is the same is at equilibrium with the rate of exit. So so when when the rate at which customers arrive into that hall.

Is is the same as the rate at which people are exiting. Then that system is said to be steady, to be in a steady state. The rate of arrival is similar to the rate of exit when it's said to be in a steady state.

OK, so if that is the case, then if you have λ arrival.

Multiplied by -1 number of customers $N - 1$. That should be the same as exit.
 N .

OK, so so if if this state is achieved, this equilibrium is achieved, then we say that this the the system is in steady state.

OK, remember P_N is a probability that there are N customers in the system.

So $n - 1$ is probably that there's $n - 1$ customers in the system.

So if you have one rate arrival, if that arrival is the same as N exit, so because μ is the exit, μ is the rate of exit. OK, so λ is the arrival rate.

And then view is the exit.

Our service rate.

Because we are assuming that once somebody is served, they exit. So the rate at which they're exiting is the rate at which they are being served. Once you're served, you have no business doing anything in.

So you exit. OK, so the rate at which they exit. So this at this state, this is what we call steady state when this equilibrium is achieved.

OK, so we can make PN the subject.

So if you make PN.

Um.

Yeah.

Let me see if we make PN the subject. We have that so I can make PN.

Uh, the subject.

And I have that so.

Now this this leads to.

This is what we call a difference equation.

Difference equation.

So that's what we call a difference equation, and we can solve that difference equation using what?

You can use. Is it? We can solve that using iteration, yeah.

Have I disappear again?



42:52

Yes.



Odhiambo, Collins 43:05

Quite unstable.

Can you guys hear me?



Straton Amodora 43:42

Yes, with an A.



Elias Shamalla 43:43

Yes, we can hear you.



Odhiambo, Collins 43:52

OK, sorry guys. Looks like today is those bad days where the network is on and off.

I'm not sure why it's appearing this way.

I'll go to check later when we take a break with the ICT guys. I don't know what's up, it's just.

It's just low disrupting us in there.

OK, uh.

Right.

Right. Let me just try to recap again. When a system is in steady state, that means the rate of arrival is the same as the rate of exit. So if this is the case.

Then if you make PN the subject you end up with this. Now this is what we call a difference equation. A difference equation can be solved by iteration. This these are very simple difference equation which you can solve by iteration.

Um.

Go a bit up here.

Right, so let's start by when N is 1.

OK, so iteration means you start with one and then you generalize 123. Then you look at the pattern and then you generalize. OK, so when N is one, I have P1.

Equals to $\lambda \mu$ is 0, so you just replace one here. See here at M.

So that's why you replace one. So you end up with with that. OK, so let's try to look at the situation when N is 2. So when N is 2.

We have P2 is equals to $\lambda \mu$.

Uh, P1.

OK, but P0P1 is equals to $\lambda \mu$.

P0.

Therefore.

P2 is the same as P is the same $\lambda \mu$.

Uh, P1, but P1 is this, so it's a lot the same as uh.

Land over me.

Which will be λ over μ .

Mm.

OK.

So what happens when P is when N is 3? When N is 3, we have P3.

But P2 is given up here. Yeah, I did.

OK, so that's P2 is giving us that. So then P3P3.

Will be the.

I love you.
Uh, Lambda.
Which?
Eventually be over me.
Great.
OK.
So who can guess me what will be?
If you look at that pattern, can I look on this for me?
If you follow the same pattern P1 is this P2.
Is this?
It.
WT5.
If you just look at it.
Someone will.
Give me a guess.
What will be 5?
Are you guys there?
Osmus.
All will be P5.
Are you able to hear me?

BO Bernard Ogol 49:34

Yes, we are able to hear you by there and I don't know that is my end alone because I can see your board is only stuck at P1.

OC Odhiambo, Collins 49:44

Oh.
Let me share it and then.
Alright, so P1 is So what what we did?
So that means there's a place that you guys didn't see. What we did is that this is the this is the equation at equilibrium or at steady state. OK, so if you make.

KR Khidir Ramadhan 50:29

The screen is still not showing.

OC **Odhiambo, Collins** 50:29

P and the subject you end up with that, so we end up with a difference equation.
Is he not showing?

KR **Khidir Ramadhan** 50:41

Yeah.

OC **Odhiambo, Collins** 50:49

OK, let me looks like so you guys are stuck where I just here.

Yes.

At P1 is that here to P1.

Let me see.

So you can't see P3, you can see P2.

KR **Khidir Ramadhan** 51:18

Yeah, so we can see the whole the the whiteboard completely.

OC **Odhiambo, Collins** 51:29

OK.

So what I can do then? Uh.

Let's let's take like a 10 minutes break. Then I can reach out to our ICT guys to see what's happening here because I'm having trouble. Let's take a short break, like 10 minutes. Then I can get in touch with these guys.

Hi everyone, I hope now it's much better.

Try to have it worked on. I'm just hoping it will work better now.

Mm-hmm.

So back to our whiteboard. I know Bernard was saying, was it?

Was that Bennett who was saying that you could not see some sections?

Uh, let's go back there and try to see if we can.

And finish up this thing.

All right, let's start again. Hopefully we can now finish up the thing that is been held holding us. So we do, we did say that.

At steady state. So let's start here again, OK?

And steady state.

So at steady state, um.

The rate of arrival is the same as.

Exit.

It should be P and minus. No, no, no, no.

OK, get rid of that. Get rid of that. So at steady state, we did say the rate of arrival.

Is is the same as exit.

OK.

And then we make if we make P_N the subject.

So you'll have $\Lambda \text{ over } \mu P_N - 1$. So this is what now we call a difference equation.

OK, and then we can solve this difference equation iteratively. The easiest way to solve a difference equation is usually iteratively. So you replace N with values until you learn the pattern. Once you know the pattern, you generalize.

Then you move on from there. OK, so we started by saying let N be one. So if N is 1, this equation here, this difference equation becomes P_1 . OK, so if N is one, it becomes P_1 is equals to Λ .

$\Lambda \text{ over } \mu P_0$.

OK, if N is 2.

If N is 2, then that equation there that difference equation becomes P_2 is equals to $\Lambda \text{ over } \mu P_1$.

OK, uh, but P_0 .

Is equals to $\Lambda \text{ over } \mu P_0$.

OK, so P_1 is $\Lambda \text{ over } \mu P_0$. Therefore if you replace this P_1 here.

Uh, with that, so you'll have P_2 .

Will become $\Lambda \text{ over } \mu$.

Um.

Of $\Lambda \text{ over } \mu P_0$.

And then that translates to $\Lambda \text{ over } \mu$.

Squared E_0 .

OK, what if N is 3?

So if N is 3, then that difference equation becomes P_3 .

Is equals to $\Lambda \text{ over } \mu$.

P_2 OK, so you just replace where there's any replace with with $3/3$ and then you end up with that which but we already know P_2 , we already know P_2 . So what it means then that P_3 .

Is equals to Λ over.

P_3P_0 because if you bring the P_2 here, if you bring this P_2 you will end up replacing it here.

End up with and that's now when I asked. I was asking Cosmos what will be if you look at that pattern, what will be P_5 ?

What will be P_5 ?

I can't hear you.

CK **Cosmas Kungu** 1:19:47

Our P_5 will be Λ over μ raised to power five times P_0 .

OC **Odhiambo, Collins** 1:19:59

Perfect. Perfect. Yeah. So, so that will be P_5 . So then P_N .

So then P_N will be Λ over μ raised to power N .

See how it's generalized. Now P_N is Λ over μ , LN uh P_0 OK.

So now we're trying to solve P_0 . What's P_0 now? The way to solve it is that P_N is a probability. Remember, remember P_N is a probability.

Um, the probability that they are N entities.

In the system.

So P_N is a probability that there are N entities in the system and because it's a probability factor, if you sum all probability you should get one. So because it's a probability if you.

If you sum all probabilities, so the invert of A is for all values of N If you sum all probabilities, you should get one.

Because P_N is a probability that there are N entities in the system, the probability that there are N customers or entities inside the system. So if you sum all probabilities, you should get.

One by the properties of probability.

OK, the properties of probability. The sum of all probabilities shouldn't give you one, but we already know P_N is this.

That so then what it means that if we sum N all values of N from zero to Infinity, remember N can only take zero to. You cannot have negative customers. It has to be either 0 customers or.

One or two or three. So if you sum all probabilities, you should get one. So if you sum all probabilities, that's the same as summing N is 0 to Infinity you sum.

λ over μ because P_N is generally λ over μ N^0 . If you sum that, you should be able to get one.

Right.

So if you sum that, you should be able to get.

to get one.

Um.

Now.

So this is the same as let's let's let's deal with this. Let's let us try to deal with this now.

OK, so let's deal with that. We can bring the P outside, can bring the P outside, P_0 outside and.

Just try to see if I can deal with.

With this.

Should be able should be equal to one. OK, so there's a power N here.

Because this is like Q . Because this is like Q , the rate at which λ λ is generally less than μ .

Because you take meal is the rate of service because it's taking long for meal to walk. You're taking long for meal.

To work to to serve if if λ the if if the rates at which so so λ the rate at which people arrive.

It has to be less than μ , so that's a fraction. This tells you that this value here is a fraction.

This is a fraction because λ is less than μ , so that's the assumption for for there to be AQ . So because it's a fraction, it's typically a fraction. Yeah, let me just say it's typically a fraction.

Uh, OK.

If you try to open up this part, so I'm just trying to sort of try to unpack what is this here. If I try to open it up like N is 0, N is 1, N is 2 up to.

Whatever. So if I try to open it up, I'll have μ power 0 plus. Remember it is a sum from zero to Infinity μ power one plus.

μ power to that way to Infinity. OK, if you if you try to unpack that expression, you'll have that to Infinity. What is this from your basic algebra? I think high school or?

First degree. What's this from your basic algebra? If what is this like?

From your basic algebra, let me just do it this way.

A.

What's that from your basic?

Is it algebra?

SW

Samwel Waiyaki 1:26:24

Is it a progression? Not sure.

OC

Odhiambo, Collins 1:26:24

What's? What's that?

Yes, perfect. That's, yeah, perfect. That's arithmetic progression. Thanks. That's that's arithmetic progression. OK, so that's arithmetic progression.

What's the sum of arithmetic progression? That's the other question, because this is arithmetic progression. Let's just see what's. How do we move on from here? So this is arithmetic progression if you want to get to the sum.

Of arithmetic progression. You say let's that be, let's call it like AR . So of arithmetic progression you multiply across with R So you have AR .

Plus AR^2 plus AR^3 plus AR^4 plus that way until you get a AR^n .

So you just multiply across across with R and then you subtract if R depending if R if R is is a ratio, it's between zero and one.

It's a ratio. Then you do you subtract a the the one on top minus the one below. If R is greater than one then you subtract you do the other way. OK, so this is just from your I think high school or whatever.

Fuel.

If you recall your arithmetic progression, OK, so for you to get the sum of arithmetic progression, this not arithmetic, it's geometric. Sorry, this arithmetic. Now this is geometric, not arithmetic. Arithmetic is where.

You have a you add there's some addition, but when the power you see you're increasing the power, the common R is being factored is the power is is increasing, then it becomes what we call a geometric, not arithmetic. So supposed to be called a geometric.

Trick.

It's a geometric progression. I think this one you may have done elementary in your high school or either your first degree introduction to mathematics or.

I don't know, but this is just very basic. It's I think it's even done in high school, high school from three or something like that.

So geometric progression. So geometric progression is. So you see the common factor A doesn't change, but R the power of R increases by a factor to some N . So if you want to sum the sum of a geometric progression.

You multiply across with R , then you subtract as long as R is a ratio. If R is a ratio or it lies between zero and one, then you subtract that expression on top to this one. So if you subtract, you'll end up with.

A lot will cancel.

Yeah, I love console.

And cause until you end up with that minus that.

And that is the same as the one on top minus R .

The one, uh, the one below.

So if you can't, if you subtract the two expressions, you'll end up with that and then now you can factor out and make a the subject. So if you factor out a , so you'll have a .

$1 - R^N$ and then yeah, you factor out a as your subject, so you have that.

Yep, Yep.

OK.

So then that's this is just basic mathematics. I don't think it should be hard for you to see. So if you make a the subject and you'll have a is equals to a .

$1 - R^N$.

Our.

one. Ah, okay.

But if N , if N tends to Infinity, R^N is sufficiently large. So if R is sufficiently large and it tends to Infinity, this value here will become zero because this is a fraction if you power a fraction to a very big.

Our it will tend to 0.

So if N tends to Infinity then then the sum of a geometric progression is just it is just this.

Where A is the coefficient that is not changing, so it's the coefficient that is not changing.

Right.

Yeah, where A is a coefficient that is not changing. So if you recall that and we try to apply here.

So this P_0 can be the coefficient that is not changing and then this this is your R , the R over μ is R and remember it N is powered to Infinity. So if you sum a geometric

progression and power you sum it to Infinity.

It is just the same as. It's just the same as one over that. So so if I do P, so if I do P_0 .

Plus P_0 .

Plus P_0 .

So if you try to unpack that expression there, you'll see something like this.

OK, so you'll have if you try to unpack this here.

Try to unpack this expression.

OK.

So if you try to unpack it from N is 0, N is 1, N is 2, so you'll have you'll have the expression appearing down here, which is a geometric expression and a geometric progression, and the sum of that geometric progression is just P_0 .

1 minus your R is Λ over μ .

As simple as that.

Yeah, so if if you come back on top here, the sum, remember the sum of probabilities here, the sum of we started here. If you sum all probabilities should get one.

But we know P_N is given as Λ over μ power $N P_0$.

It's given us that P_0 .

Yeah.

Right. I hope it's making sense. Uh.

Uh.

OK, so if you sum these, you should get one. But we have seen that this part here, this part here is a geometric progression. It's a geometric progression and it's end up with.

It's a geometric progression and you can write it in this form. So what that means then is that P_0 .

All over 1 minus that.

Should give you one.

OK, therefore what that means then that P_0 is just one minus.

I'm down by me.

OK, so.

So in terms of practical application of this P_0 , this we say that P_N is the probability there is N entities in the system and so P_0 is a probability that there is 0 entities, there is no person, no customer in the system.

So in terms of interpretation, this is the probability that the server is idle is not doing any work.

That's the probability that the server is not doing any work. It is basically $1 - \mu$. What this means then is that probability that there's at least one person in the system.

Is just λ / μ , which is what we call that the probability that the server is busy.

The probability that is.

Right. So this is the probability that there are no customers in the system or probability the server is idle.

These are some of the performance measure of a system that we'll be interested with when we are doing modelling and of course this one is now the probability that the server.

Probability that this that uh uh like this our.

Um.

The server is busy.

It's these are some of the performance measures that that we'll be looking at in a practical system.

So we've not, we've already done the probability part. Let's look at the mean, the expected number of entities in the system.

OK.

Expectants.

Number of entities.

OK, so the expected number of entities in the system. So then they they expect a number of entities as the mathematical expectations. So $E[N]$.

Should be.

The sum for all values of N .

So it should be the value for the four values of N is P_N . So this is the same as the sum N is equals to 0 to Infinity of $N P_N$.

P_N .

And then this is the same as the sum.

And it goes to 0.

So P_N . We found P_N to be N .

And then P_0 , but P_0 we found to be $1 - \mu$.

OK.

So now we try to solve.

This so try to unpack and solve this expression so we can bring the. Usually when you

have this type of expression, the best way is to remove anything that is not relevant, put it outside the the summation.

Or integration if you're dealing with integration. So in terms of summation you pull it, so you have one.

And now.

OK, so uh, this is the.

This is the the expectation or the expected number of entities inside this system.

So if you let's try to work on this part here, try to work just on this part here, try to unpack.

That part there.

So if you start with 0, any 00, so you'll have zero plus if you start, if your N goes to one, you'll have 11, you'll have that.

OK.

And then if you go to any two, you'll have two here.

Lambda over.

You have that power two. If N is 3, you have three here, Lambda over MU power three, so that way.

OK.

What does this look like? I don't know if any of you have an experience with some of this advanced algebra.

Does this looks like this looks like uh huh?

$1 + R + 1 + R + 2 R^2 + 3.$

$+4 R^4$ up to some NRN.

Somebody with an idea. Let me just try to test. Is it? This should be algebra.

Try to test your algebra. Anyone with idea what is that?

Leonard, can you think of that?

This time you are not getting line. Somebody else has to get it.

BO Bernard Ogol 1:44:02

Yeah, sure. OK. No, I think.

OC Odhiambo, Collins 1:44:04

What? What's that?

BO Bernard Ogol 1:44:07

The children increasing.

The metric progression.

 **Odhiambo, Collins** 1:44:11

Yeah.

Yes, thanks. At least. But this time we're not getting lines. If it was somebody, you would have gone in. But that's perfect, Bernard. This is an increasing geometric progression.

It's an increasing what we call increasing geometric progression because you see the coefficient is increasing with the power of the common ratio. So if you have the coefficients that is increasing with the power common ratio.

When we talk of increasing geometric expression.

Right. And there's a way we get the mean.

Of that. So there's a way we work on the mean of that. Um.

For you to get the mean, there's a way we work on that. So we can say this is AR , so we multiply across with R or we divide across with R whichever way. So we can multiply across with R so that we have R here plus.

Plus $R^2 + 2R + 3R^4 + 4R^5$. So if you multiply across with R .

OK.

You have a RR . If you multiply across with R and then you subtract the one on top minus the one below, assuming that your R , which is a common ratio, it's actually a ratio between zero and one.

I.

So if you subtract and then if you multiply across with R , then you subtract that minus that, you'll end up with $1 - NR$.

$N - NR$ $R + 1$ and then this side you'll end up with AR .

Oh.

But what happens if you subtract, you get you have $1 -$ minus, then the rest will go.

OK, so if you subtract, you'll not get that whatever I'm trying to put there.

You don't get that. You get a geometric.

Let me just cancel this faster, faster. So if you subtract the two this side on top you'll end up with let me see if you subtract 1 .

Um, I will disappear.

Should eventually give you.

A geometric progression. Let me see if you subtract that and start because that was

supposed to be one. OK, yeah, so if you subtract, you'll end up with another geometric progression here.

Up to.

Yeah, so if you subtract, you'll end up with another geometric progression here, so that this is just a geometric progression.

Is equals to if N is.

Like so N remember the the limits as N tends to Infinity so so that that will give you if N tends to Infinity for.

For anything that is ratio that is between zero and one, it should give you 0. So if you subtract you should end up with AR.

And pays out.

Therefore your AR.

One, one, one.

Point.

OK, so that tells you that.

If we if you try to unpack this this expression here, you'll end up with a geometric progression and the sum of a geometric progression is basically one of 1 minus the common ratio squared as long as.

You are summing to Infinity.

As long as you summing to infinity.

OK.

Hope it's not too much maths there. It's just.

You must have done in your high school or mathematics introduction to mathematics terminal or yeah, just basically I tried to solve my.

I tried to solve now this part here.

So this part here will be let me write it down.

But try to solve that expectation of N .

Is, uh, this thumb?

Is N .

OK, um, so if I bring the common factor out.

I bring out the common factor I have zero and.

Wow.

I have Λ over μ .

And.

OK so if I try to if I try to open the bracket, if I try to open the bracket I will have so

new and the outside I have that.

Then inside if I try to open the bracket I have zero and then I have one. So if N is 0 we have 0 plus when N is one we have Λ over μ .

One you.

When N is 2, we have two Λ over μ^2 .

When N is 3, so when N is 3 we have 3 Λ over μ^3 .

When N is small, we have four Λ over μ .

That way and then it continues to Infinity. OK, so just reorganizing this expression I have.

Um.

I can factor out to make this a geometric I can factor out.

ΛW remember geometric has to start from from 1:00 so that I have that.

From the.

Just.

Um.

OK.

OK, so this part here.

That part here is a progressive geometric. It's an increasing geometric progression.

It's an increasing geometric progression. And if you sum an increasing geometric progression is the same as getting 1 minus.

They are the common ratio where where in this case is is Λ over μ .

But now you square.

Square OK because it's a geometric progression, so it's is that then in square.

Yeah.

Right. Too much mathematics. Let's see how we can finish wrong.

Uh.

So now we have the E expectation of N as being.

1 minus.

OK, that make make it Λ over μ .

So there was that and then then there's this, this whole chunk. Yeah, that is just one minutes Λ one.

But this is squared.

OK.

So now you can organize this, you can rearrange. So one will cancel, one will cancel, then reorganize it so that rearrange your. So this one will cancel down here. This will

cancel with one of this.

And then now you have that that uh.

Then you have. Then you'll have hand over meal.

Multiplied by.

Is it no?

Like if you let me just do it.

We divide by that, so that's it's like.

That's like μ minus, so they take μ on top.

So that's now you have that is just basic algebra. Just rearrange those numbers, those those values, just to rearrange the values.

So that this is the average entities in the system.

That's the average entity synthesis. What this means then is that if I know the rate of arrival, if I know the arrival rate and I know the service mechanism, I can know how many now on average how many.

People are in the system. I can get on ours.

As long as those assumptions that we stated for an MM one model are as long as those assumptions hold.

I can and consider MM1 model where arrival is random. There's only one server, there's no bulking, then service mechanism. There's only one server, so my service mechanism will follow exponential.

As long as I know the value of μ , which is the service mechanism rate and I know the arrival rate, I can I can come up with the average number of entities that are in the system.

OK, so later on we'll do the application of it. Right now I wanted you to see the mathematics. So and I've chose a very basic model. As I told you, if you want to grow further then it becomes a bit complex.

But now we've just developed a very simple model or giving model and then we're trying to come up with performance measure of that of that model. So one interesting thing is the average number of clients in the system.

Uh, the other interesting thing is this is the average entities in the system.

OK.

Something another performance measure of interest will be the average entities in the queue.

The other thing is the average entities in the queue. The average entities in the queue is $n - 1$ is the expectation of $E_n - 1$ because the person who is being served.

At the counter is not considered to be in the queue, so so not everybody in the system are in the queue.

And we are seeing it only. It only has one server, so the person who's been served is not on the queue.

OK, so you do $N - 1$ and to get that. So the other the other bits are the same, so the mathematical expectation of $N - 1$. So this time you start with one because you can't have a negative pass one.

Then you have $N - 1$ and then you have PN . So this is the sort of thing we are trying to evaluate now.

So if that's the case, then we already know our PN , so we'll just try to evaluate this.

$N - 1$ PN was 1 minus that.

And.

That over that N OK, remember $PNPN$ was that.

And P_0 but P_0 we are we we we have already shown P_0 is basically this.

I think P_0 is that, so that's why PN is given us that.

Right, so now trying to unpack what is this, we can open up the the bracket to solve this.

Uh-huh. So the easiest ways to get everything without an end outside.

Then and then so.

And then we deal with $N - 1$.

-1 .

And we deal with that.

Alright, uh.

This this will give us um.

One minus this will give us.

And give us an.

Oh.

And by the way, there are many other ways probably you may find.

You may find a an easy way, I don't know.

Yeah, different ways of solving these.

Uh, this algebraic expression.

I'm just showing you the the way I think it's it's easier for me, but with the current age you may find an easy way of solving it mathematically. Yeah, you may find another different way in.

But for those people do.

I'll do mathematical what we call actuarial science, actuarial mathematics. We always encourage them to come up with another method that is called mathematical reasoning.

So you can come up with what we call general reason.

And and and and look at one person is not in the queue.

Then you can apply general reasoning and avoid the mathematical description the way we are doing, but you can apply general reasoning and eventually end up with the same results that we are going to get.

For example, if this is the average entities in the queue in the system, then what if we subtract one person who is being served? So you subtract, let's say the server is not busy P1.

Then you can use mathematical reasoning to end up with what we are going to get, which will eventually get this give you the same results with what we are going to do.

OK, so there are people who use general reasoning to to solve.

But that one we use it a lot in finance anyway, for those who are doing absolute science.

For us here, this is a mathematical way of doing it.

This there's a way you can manipulate this because this is this first part. This is a geometric progression.

Um, this is a normal.

Progression. No normal geometric progression, but this is an increasing geometric progression. So you have this. You can work on those two paragraph paragraphs there, OK.

So try to see that. Let me give it as an exercise and give it to be part of our of our assignment as an exercise to show that you'll eventually end up with.

Uh, let's see.

Yeah.

You end up with that.

That would be OK to show that you'll eventually end up with that.

Let me take you back just to verify that's this correct position.

So somebody will ask me uh.

What? What does this have to do with data science?

What does the this whole expression has to do with data science?

Very interesting question that you may ask. The answer is that because it is.

The answer is that because it's part of the stochastic process, you will definitely at

some point or later, depending on where you work, you may come up with a stochastic you may end up requiring.

Some knowledge in stochastic processes when you are modelling, when you are analysing or predicting some data, but you need to incorporate the stochastic nature of your data. So if you need to, if you need to incorporate the stochastic nature of your data.

Then you need to have stochastic reasoning. You need to have a stochastic kind of reasoning so that you know in stochastic data is not conventional the way you probably would know it. Yes, there is data, but that data is ordered with some parameter space.

OK, there's some parameter space, and so the nature of your model should include aspects of that parameter space.

The good aspects.

That's.

OK, for parameter space. Uh.

And of course.

Of course, the mathematics is helpful for you to have a theoretical background background of of it.

You may say I was. I was listening to a student who who was asking me.

Can I do data science?

And and avoid the mathematics as much as possible. The student was doing masters in data science at at one of the universities here. So it was like which which I wanted data science.

Course master's degree in data science, but I want a data science that has analysis, but there's no mathematics. So I was like my friend, you can't avoid some of these things because one of the.

Whereas we agree with you that you want the knowledge in order for you to.

You want the knowledge in order for you to apply it to where you work. We also need to prepare you for your future advanced course. Like for example, if you you are to come back for a PhD, the PhD will assume the course you'll do will assume that you have a.

A good theoretical background, not just throwing data inside there, running the scripts and that's it. But do you have a mathematical or do you have a concrete, a solid background and and ground in terms of what you're applying?

OK, so this is important for your theoretical formation so that you have a a solid

background.

Eventually you may not really need the formula in application, but you need it in understanding how the data concept, how the data flows and how to interpret your data. OK, so you need you need to understand how you'll eventually.

Interpret, interpret your data.

So we talked of steady state here and here typically a system is in steady state when this is true.

And solving this difference equation iteratively, we start with one. If you put N to be one, you end up with that. If you put N to be two, you end up with that, and then you replace, you end up with that. So if you continue the same way, you'll end, you'll end up with.

With that. So we've seen that from our whiteboard.

And now we try to solve for P So this one I just said like here let you see we let this one $\lambda \mu$ be row. So I've let $\lambda \mu$ to be row.

So that instead of $\lambda \mu$ I work with ρ in this case. So if I do this and then you sum the sum of all probabilities should be one. If you sum this should get one and then this is a geometric progression.

And because the geometric progression and that row is a ratio between zero and one, a geometric progression, the sum of a geometric progression is $1 / (1 - R)$ or $1 / (1 - \rho)$. So this is the sum of a geometric progression.

OK, so then if you replace it and then eventually come and replace ρ with $\lambda \mu$, you'll end up with what we've found.

Um.

The next part we were doing now average entities in this system, average entities in this system. So you have N , so we've talked of this.

So we've talked of this. So here I've just replaced ρ with $\lambda \mu$ just to try to make it much easier. So, so this, this is an increasing geometric progression and increasing geometric progression will end up with that.

That and then outside we add that. So if you replace, you'll end up with that.

For the average entities that is in the queue, we do $N - 1$, so we do $N - 1$ and then you'll eventually end up with that. So this one I'll leave it for you.

You'll end up with that as your final answer.

To be part like, yeah, I've actually solved it. I don't think I should deal and put it, but I've actually solved it. But I'll share with it. I'll share with it. It will be part of your assignment.

And then as we find finalize is what we call the little slow. So this little slow gives the relationship between average number of customers and the waiting time in the system vis-a-vis also.

Average cost, uh, average cost is a be the waiting time in the system.

So the waiting time in the system is basically λ times W and W is already W the waiting time in the system. So if you make W the subject, you'll end up.

With that and if you make the subject to end up with that.

All right, too much mathematics today. I think we can't stop there to to make us absorb.

That's.

Do you have any question?

Right. Uh, I don't know if the mats have been.

I've been too much than expected. It's the nature of the course. I'll try next few weeks to make sure we have very little months, but just to make sure you also capture what you are supposed to capture at the end of it. It's also been just easy.

At the end of it, you finish this course and when you come for your PhD and you are required to have a bit of stochastic and you have no idea, so it's so so it will be injustice. So it's so I'm trying to also solve that.

Apart from try to reduce so like for the.

Or that other other stochastic processes like branching processes, death and birth and death processes. I'll try as much as possible to see if I can reduce as much much so that we for even this one I've just picked.

The least, the most easiest or least model, stochastic model. There are those ones that are a bit advanced. So those ones have left it for the gurus who will come back for your PhD.

And uh, PhD at Strathmore, I think that we were coming up with a taught program where you are taught for an year.

We make sure you're grounded before even you start your thesis. If your thesis is in survival analysis or is in predictive modeling some some area, we make sure you're grounded in those theoretical skills before you can actually.

Um, sit down and do your thesis.

OK. So I think we can stop there just to give you guys a break and enjoy the rest of your evening.

CK **Cosmas Kungu** 2:16:48
Thank you.

SA **Straton Amodora** 2:16:50
Thank you.

SK **Stephen Korir** 2:16:51
Thank you. Thank you.

● stopped transcription